

Amrit Gopinath

Undergraduate Researcher – NLP & Multilingual Representation Learning

CONTACT

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EDUCATION

Sri Sivasubramaniya Nadar
College of Engineering, Chennai
B.E. Computer Science and Engineering
2024 – Present
CGPA: **9.383/10.0** (4th sem)
Ranked **1st** in Department

RESEARCH INTERESTS

Natural Language Processing
Multilingual Representation Learning
Mechanistic Interpretability
Mixture-of-Experts LMs
Classical Tamil NLP
Synthetic Media Detection

TECHNICAL SKILLS

Programming
Python, C, TypeScript, Java, LaTeX, Git
Machine Learning
PyTorch, Transformers, Transfer Learning
NLP
Representation Learning, Probing, MoE
Research Practice
Shared Tasks, Ablation Studies, Error Analysis

SELECTED ACHIEVEMENTS

- Consistent shared-task participant across MediaEval, CLEF, and FIRE (2025–2026).
- FinMemeval 2026** – Top-10 in English, Chinese, and Arabic tracks.
- Merit Scholarship** – 3rd Position, First Year.
- Top 10** – GDG Hackathon.

ACADEMIC SERVICE

Reviewer / Sub-reviewer
CLEF 2026 Working Notes

RESEARCH PAPERS

A Declarative–Procedural Perspective on Expert Routing in Bilingual MoE Language Models 2026

ARR-Submission & Target venue: BlackboxNLP | Arxiv Preprint | First author

- Analyzed how bilingual English-German MoE Transformers route experts by language and linguistic structure, using probing and information-theoretic analysis.
- Led the research design, experimentation, and writing.

Exploring Swin Transformers, Luminance Input, and Hybrid Architectures for Synthetic Image Detection 2026

MediaEval 2026 | Working Notes | First author

- Compared Swin Transformer, luminance-only, and hybrid CNN architectures for detecting AI-generated images and best F1 score was **0.6997**.

FinMMEval 2026 Shared Task Submissions – TextSentinels 2026

CLEF 2026 | Working Notes – Accepted | Co-author

- Task 1 – Multilingual Routed RAG for Financial MCQA:** FAISS retrieval with an LLM router and scoring ensemble across English, Chinese, and Arabic; placed **6th** (English), **6th** (Chinese), **7th** (Arabic).
- Task 2 – BM25-Grounded Multilingual Financial QA:** pipeline over SEC filings and tabular disclosures, local ROUGE-1 F1 improved 0.04 to 0.28 (0.18 on the final benchmark).
- Task 3 – Multi-Asset Financial Decision Making:** iterative trading agent, from a FinBERT sentiment pipeline to an XGBoost classifier with a sentiment-memory feature; **1.14** cumulative return, **2.03** Sharpe ratio in live forward-testing.

MediaEval 2025 Working Notes: Synthetic Image Detection and Segmentation 2025

MediaEval 2025 | Working Notes | First author

- Built systems to distinguish AI-generated from authentic imagery in a shared-task setting.

RESEARCH WORK

Dravidian-CodeMix Shared Task – Tamil Hate Speech and Tulu Hope Speech Detection 2026

FIRE | Team member

- Built two IndicBERT-based systems: hierarchical hate-speech classification for Tamil, and joint coarse/fine-grained hope-speech classification for Tulu.
- Explored encoder choice and layer-unfreezing strategies to optimize each task pipeline.

RESEARCH EXPERIENCE

NLP Research Intern 2026 – Present

National Institute of Technology Tiruchirappalli

- Researching representation learning and evaluation of neural language models, including sentence representations and multilingual probing.
- Studying classical Tamil computationally – clause structure, predicates, and syntax-aware retrieval.

RESEARCH-TECHNICAL PROJECTS

A-Eye – Assistive Vision System

- Built a dual-camera prototype combining object detection and audio feedback for real-time spatial awareness.
- Received institutional research funding; showcased at Innovation Day 2025, SSN College of Engineering.